

Data Management Goals

- 01** | Understanding and supporting the information needs of the enterprise and its stakeholders, including customers, employees, and business partners
- 02** | Capturing, storing, protecting, and ensuring the integrity of data assets
- 03** | Ensuring the quality of data and information
- 04** | Ensuring the privacy and confidentiality of stakeholder data
- 05** | Preventing unauthorized or inappropriate access, manipulation, or use of data and information
- 06** | Ensuring data can be used effectively to add value to the enterprise

Section 2 — Essential Concepts: Overview

- 01 | 2.1 Data — what data is, representation, context, and the disciplines required to manage it
- 02 | 2.2 Data and Information — the DIKW pyramid, its limitations, and why both data and information must be managed
- 03 | 2.3 Data as an Organizational Asset — data as economic resource; value, monetization, and the data-driven organisation
- 04 | 2.4 Data Management Principles — 12 foundational principles grouped in four themes
- 05 | 2.5 Data Management Challenges — 13 challenges that arise from the properties and principles of data
- 06 | 2.6 Data Management Strategy — strategy components, deliverables, and the CDO's role

What Data Is

Nature of Data

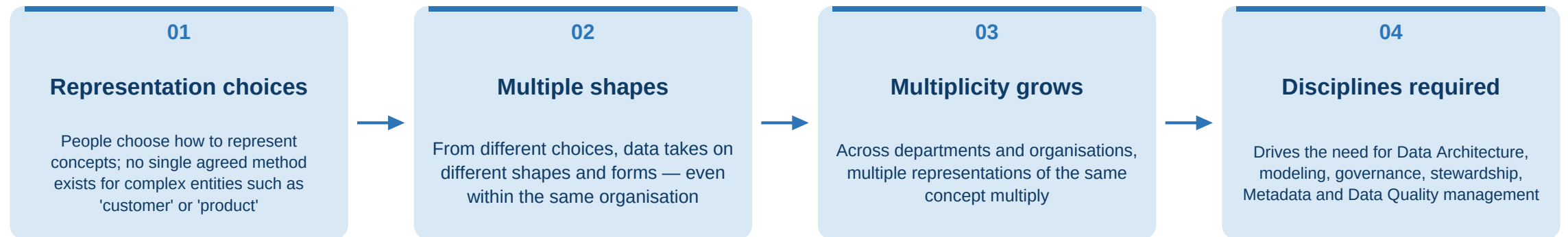
- Represents facts about the world; in IT, also understood as information stored in digital form — though data is not limited to digitised information
- A means of representation: 'stands for things other than itself' (Chisholm, 2010)
- Both an interpretation of objects it represents and an object that must itself be interpreted (Sebastian-Coleman, 2013)
- Context = data's representational system; the conventions of that system are documented as Metadata

Why Management is Needed

- People make different choices about how to represent concepts, creating multiple shapes for the same idea
- Even within one organisation, different departments represent the same concept differently
- Across organisations, the problem of multiplicity multiplies, requiring industry-level data standards
- Multiplicity drives the need for Data Architecture, modeling, governance, stewardship, and Metadata and Data Quality management

Why Data Representation is Complex

Different choices about how to represent concepts create compounding challenges that require dedicated management disciplines.



Data and Information

The DIKW Pyramid

- Data is called the 'raw material of information'; information is called 'data in context'
- A layered pyramid is often used: Data → Information → Knowledge → Wisdom
- Useful for explaining why data needs to be well managed
- Throughout the DMBOK, 'data' and 'information' are used interchangeably

Three Challenges with the Pyramid

- It assumes data simply exists — but data has to be created
- Its linear sequence fails to recognise that it takes knowledge to create data in the first place
- It implies data and information are separate — in reality they are intertwined and dependent on each other
- Central tenet: BOTH data and information need to be managed together with uses and customer requirements in mind